

The British Sector of the Western Front, 1914–1918

The Historic Environment · Injuries, Treatment and the Trenches · Complete Mastery & Source Studio

Pupil Name: _____

Class / Set: _____

Target Grade: []

SECTION A PROGRESS & ASSESSMENT TRACKER (16 MARKS TOTAL)

Marked by Teacher or Peer Verified

Assessment Component & Stem Focus	Format Style	Max Marks	Pupil Score
 Complete Western Front Knowledge Vault (All 100 Recall Questions)	Checklist	/100	_____
Round 1 (Stepped): Q1(a) Feature 1 (Casualty Clearing Stations)	Stepped Ladder	/2	_____
Round 1 (Stepped): Q1(b) Feature 2 (Thomas Splint)	Stepped Ladder	/2	_____
Round 1 (Stepped): Q2(b) 4-Part Follow-Up Grid (Base Hospital Water)	Stepped Ladder	/4	_____
Round 1 (Stepped): Q2(a) Source Utility (Somme CCS vs Base Hospital)	Stepped Ladder	/8	_____
Round 2 (Dual-Track): Q1(a) & Q1(b) Features (Mobile X-rays & Trench Fever)	Dual Track	/4	_____
Round 2 (Dual-Track): Q2(b) 4-Part Follow-Up Grid (Gas Casualties)	Dual Track	/4	_____
Round 2 (Dual-Track): Q2(a) Source Utility (RAMC Logbook & Flanders Mud)	Dual Track	/8	_____
Round 3 (Simulation): Full Section A Timed Exam Pitch (Q1a, Q1b, Q2a, Q2b)	Exam Hall (25m)	/16	_____
TOTAL COMBINED EXAM PRACTICE MARKS:		/48	_____ / 48

Teacher / Peer Marker: _____

Date: _____

Overall Grade: [9 8 7 6 5 4]

SECTION A EDEXCEL EXAM FAST FACTS & STEM BLUEPRINTS (16 MARKS · ~25 MINS)

OPTION 11 PAPER 1 SECTION A

- **Q1(a) & Q1(b) Features (2 Marks each · ~3 mins each):** *F-D Formula*. State **Feature** in sentence 1 → support with 1 **precise supporting factual detail**. (2 separate questions).
- **Q2(a) Source Utility (8 Marks · ~12 mins):** *C-O-P Matrix*. Evaluate both sources for **Content** (what it says) → **Own Contextual Knowledge** → **Provenance** (Nature, Origin, Purpose / NOP).
- **Q2(b) Follow-Up Investigation (4 Marks · ~7 mins):** *4-Step Enquiry*. Quote **Detail in Source B** → Formulate **Targeted Question** → Name **Specific Historical Source Type** → Explain **Direct Purpose**.

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Complete Knowledge Retrieval Vault · Questions 1 to 50 (Trench Environment, Ill-Health & Wounds)

VAULT PART 1

☐ 1. By 1900, how had surgical practice in civilian hospitals evolved from Joseph Lister's earlier methods? [KT5.1]	☐ 2. Which of the following was a key piece of technology used in aseptic surgery by the outbreak of the First World War? [KT5.1]
☐ 3. In what year did the German physicist Wilhelm Roentgen accidentally discover X-rays? [KT5.1]	☐ 4. Which of the following was a major limitation of using early X-ray machines in 1914? [KT5.1]
☐ 5. Why were early X-ray machines highly unsuitable for the fast-paced, mobile environment of a battlefield in 1914? [KT5.1]	☐ 6. In 1901, the Austrian doctor Karl Landsteiner made a breakthrough that revolutionized blood transfusions. What did he discover? [KT5.1]
☐ 7. Before the discovery of sodium citrate in 1915, what was the massive problem with performing blood transfusions on the battlefield? [KT5.1]	☐ 8. When the British Expeditionary Force (BEF) arrived on the Western Front in 1914, what geographical feature made the medical situation catastrophic? [KT5.1]
☐ 9. What is the military definition of a 'salient', such as the famous one at Ypres? [KT5.1]	☐ 10. Which major British offensive, launched in 1916, became a medical disaster with over 57,000 casualties on the very first day? [KT5.1]
☐ 11. What unique medical facility was established during the Battle of Arras in 1917? [KT5.1]	☐ 12. During the Battle of Cambrai in 1917, what groundbreaking medical logistical system was used for the first time by Oswald Robertson? [KT5.1]
☐ 13. Why did aseptic surgery (sterile operating theatres) prove to be nearly useless in frontline dressing stations in 1914? [KT5.1]	☐ 14. What made transporting wounded soldiers away from the frontline so incredibly difficult? [KT5.1]
☐ 15. To what extent was the Royal Army Medical Corps (RAMC) prepared for the medical challenges of the Western Front in 1914? [KT5.1]	☐ 16. What did the term 'war of attrition' mean for the soldiers living in the trench system? [KT5.1]
☐ 17. Why were horse-drawn ambulances often ineffective in the worst conditions of the Western Front? [KT5.1]	☐ 18. Which of the following best describes the layout of the trench system on the Western Front? [KT5.1]
☐ 19. In what year did Karl Landsteiner discover that blood group O was the universal donor group? [KT5.1]	☐ 20. Which statement accurately summarises the 'historical context' of medicine at the outbreak of war in 1914? [KT5.1]
☐ 21. What was the direct environmental cause of 'Trench Foot' on the Western Front? [KT5.2]	☐ 22. What were the symptoms of severe, untreated Trench Foot? [KT5.2]
☐ 23. How did the British Army attempt to clinically prevent the spread of Trench Foot in permanently flooded trenches? [KT5.2]	☐ 24. What was 'Trench Fever', and what were its primary symptoms? [KT5.2]
☐ 25. It took years for medical officers to discover the cause of Trench Fever. What was eventually found to be the transmitter? [KT5.2]	☐ 26. How did the British Army strategically adapt to combat the epidemic of Trench Fever once its cause was discovered? [KT5.2]
☐ 27. What was the 'Ypres Salient'? [KT5.2]	☐ 28. Why was the First Battle of Ypres (Autumn 1914) strategically critical for the British? [KT5.2]
☐ 29. What terrifying new weapon was first used by the German Army at the Second Battle of Ypres in 1915? [KT5.2]	☐ 30. During the fight for Hill 60 in April 1915, what tactic did the British successfully use to defeat the German positions? [KT5.2]
☐ 31. Why did the British launch the massive Battle of the Somme in 1916? [KT5.2]	☐ 32. What made the Battle of the Somme a 'medical nightmare' on its very first day? [KT5.2]
☐ 33. Because the ground at Arras was chalky and easy to tunnel through, what incredible feat did British and New Zealand miners achieve in 1917? [KT5.2]	☐ 34. In the organisation of the trench system, what was the primary purpose of the 'support trench', located roughly 80 metres behind the frontline? [KT5.2]
☐ 35. Why were trenches deliberately dug in a zig-zag pattern instead of a straight line? [KT5.2]	☐ 36. How did the complex, zig-zag design of the trenches unintentionally create severe problems for the medical services? [KT5.2]
☐ 37. Why did the army have to rely heavily on horse-drawn ambulance wagons instead of modern motor ambulances near the frontline? [KT5.2]	☐ 38. What was the main medical danger caused by delays in evacuating wounded soldiers across the devastated, muddy terrain? [KT5.2]
☐ 39. What was 'Dysentery' and how was it spread on the Western Front? [KT5.2]	☐ 40. What does the medical condition 'Shell Shock' (now understood as PTSD) reveal about the environment of the Western Front? [KT5.2]
☐ 41. Which type of weapon was responsible for the highest percentage of wounds (roughly 58%) on the Western Front? [KT5.3]	☐ 42. Why were deep shrapnel wounds almost guaranteed to become severely infected within hours? [KT5.3]
☐ 43. What was biologically unique about the mud in Flanders and France that made it so deadly when dragged into a wound? [KT5.3]	☐ 44. What is the defining, fatal characteristic of the 'Gas Gangrene' infection? [KT5.3]
☐ 45. Why did traditional 19th-century surgical antiseptics (like carbolic acid spray) fail to cure Gas Gangrene on the Western Front? [KT5.3]	☐ 46. How did the British Army successfully reduce the threat of Tetanus infections on the Western Front? [KT5.3]
☐ 47. At the start of the war in 1914, why were head injuries disproportionately high (accounting for roughly 20% of wounds)? [KT5.3]	☐ 48. What crucial piece of protective equipment was introduced in 1915, drastically reducing fatal head injuries by 80%? [KT5.3]
☐ 49. What terrifying new weapon did the German army introduce at the Second Battle of Ypres in April 1915? [KT5.3]	☐ 50. What was the physical effect of a Chlorine Gas attack on a soldier? [KT5.3]

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Complete Knowledge Retrieval Vault · Questions 51 to 100 (Chain of Evacuation & Medical Advances)

VAULT PART 2

☐ 51. Before official gas masks were widely issued in July 1915, how did soldiers desperately attempt to survive gas attacks? [KT5.3]	☐ 52. Which poisonous gas, introduced later in 1915, was significantly faster-acting than chlorine and capable of killing an exposed person within two days? [KT5.3]
☐ 53. What made Mustard Gas (introduced in 1917) such a horrific and distinct weapon? [KT5.3]	☐ 54. Despite the immense psychological terror it caused, why did poison gas actually account for under 5% of all British deaths (around 6,000 soldiers)? [KT5.3]
☐ 55. Which debilitating environmental illness was spread by the millions of body lice living in the seams of the soldiers' unwashed uniforms? [KT5.3]	☐ 56. How did the British Army attempt to medically prevent the spread of Trench Foot in the flooded trenches? [KT5.3]
☐ 57. What was 'Shellshock'? [KT5.3]	☐ 58. Which specialist medical facility in Edinburgh was set up specifically to treat British officers suffering from severe Shellshock? [KT5.3]
☐ 59. What exact acronym did the British Army officially use in medical records to categorise cases of Shellshock? [KT5.3]	☐ 60. Why did the brutal environment of the trenches force the Royal Army Medical Corps (RAMC) to dedicate huge resources to 'preventative measures'? [KT5.3]
☐ 61. What was the fundamental purpose of the 'Chain of Evacuation' on the Western Front? [KT5.4]	☐ 62. Who had the dangerous task of initially rescuing wounded men directly from No Man's Land under enemy fire? [KT5.4]
☐ 63. Where was the Regimental Aid Post (RAP) physically located? [KT5.4]	☐ 64. What was the medical capability of the Regimental Medical Officer stationed at the RAP? [KT5.4]
☐ 65. Who staffed the Advanced Dressing Stations (ADS) and Main Dressing Stations (MDS)? [KT5.4]	☐ 66. How did the British military respond to the violent, injury-worsening journeys caused by horse-drawn ambulance wagons? [KT5.4]
☐ 67. Why did the RAMC still have to rely heavily on horse-drawn wagons (sometimes needing six horses) even after acquiring motor ambulances? [KT5.4]	☐ 68. How were massive numbers of stabilized casualties (up to 800 at a time) transported from the CCS to the Base Hospitals? [KT5.4]
☐ 69. Where were the Casualty Clearing Stations (CCS) strategically located? [KT5.4]	☐ 70. What vital medical sorting process took place at the Casualty Clearing Station (CCS)? [KT5.4]
☐ 71. Because deadly gas gangrene developed so rapidly, how did the role of the Casualty Clearing Station (CCS) critically change during the war? [KT5.4]	☐ 72. Where were Base Hospitals located on the Western Front? [KT5.4]
☐ 73. As the war progressed and the CCS took over emergency amputations, what did Base Hospitals begin to specialize in? [KT5.4]	☐ 74. To cope with the unprecedented scale of casualties, how drastically did the RAMC (Royal Army Medical Corps) expand between 1914 and 1918? [KT5.4]
☐ 75. Who were the QAIMNS, whose numbers grew from 300 to 10,000 during the war? [KT5.4]	☐ 76. What does the acronym FANY stand for? [KT5.4]
☐ 77. What was the most famous and physically demanding frontline role undertaken by the female volunteers of the FANY? [KT5.4]	☐ 78. Aside from driving ambulances, what other vital morale and hygiene services did the FANY provide? [KT5.4]
☐ 79. What was incredibly unique about 'Thompson's Cave', built by British and Commonwealth troops in 1916? [KT5.4]	☐ 80. Why was the underground hospital at Arras highly significant for treating casualties? [KT5.4]
☐ 81. Why was traditional 19th-century aseptic surgery effectively impossible to maintain on the Western Front? [KT5.5]	☐ 82. Faced with rampant gas gangrene, surgeons began using a technique called 'Debridement' (Wound Excision). What did this involve? [KT5.5]
☐ 83. What was the 'Carrel-Dakin method' used to treat deep infections? [KT5.5]	☐ 84. What was the main logistical difficulty with the highly effective Carrel-Dakin solution? [KT5.5]
☐ 85. If antiseptics, debridement, and the Carrel-Dakin method all failed to stop the rapid spread of gas gangrene, what was a surgeon's only remaining option to save the patient's life? [KT5.5]	☐ 86. Before 1915, what was the estimated mortality (death) rate for a soldier who suffered a fractured femur (thigh bone) from a gunshot or shrapnel wound? [KT5.5]
☐ 87. Why did a fractured femur cause such a high death rate during early transport? [KT5.5]	☐ 88. Who originally designed the Thomas Splint before the outbreak of the First World War? [KT5.5]
☐ 89. How did the Thomas Splint (introduced to the Western Front by Robert Jones in December 1915) miraculously increase the survival rate for thigh fractures from 20% to 82%? [KT5.5]	☐ 90. What diagnostic technology was absolutely essential for accurately locating deeply embedded shrapnel so surgeons could remove it before infection set in? [KT5.5]
☐ 91. How was X-Ray technology adapted to deal with the vast number of casualties closer to the frontline? [KT5.5]	☐ 92. What was a major technical limitation of the early mobile X-ray units used on the Western Front? [KT5.5]
☐ 93. Early blood transfusions (such as those pioneered by Dr Lawrence Bruce Robertson) were heavily limited because blood could not be stored. How did this force doctors to perform transfusions? [KT5.5]	☐ 94. In 1915, what chemical did Richard Lewisohn discover could be added to blood to successfully stop it from clotting? [KT5.5]
☐ 95. Following Lewisohn's discovery, Richard Weil discovered that blood mixed with sodium citrate could be refrigerated for up to two days. What monumentally significant breakthrough did Francis Rous and James Turner make in 1916? [KT5.5]	☐ 96. Which American doctor utilized these chemical discoveries to establish the world's first 'blood depot' (blood bank) before the Battle of Cambrai in 1917? [KT5.5]
☐ 97. How did Oswald Hope Robertson successfully store the 22 units of pre-donated universal Type O blood for his pioneering blood bank at Cambrai? [KT5.5]	☐ 98. Which pioneering surgeon developed early plastic surgery techniques to rebuild the severe, disfiguring facial injuries caused by shrapnel? [KT5.5]
☐ 99. Why did the brutal environment and massive casualty numbers of the Western Front force rapid medical advancements? [KT5.5]	☐ 100. Which of the following represents a major diagnostic limitation of the mobile X-ray units? [KT5.5]

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Official Mark Scheme & Knowledge Vault Answers · Answers 1 to 50 (Self & Peer Marking Bank)

MARK SCHEME 1

💡 Quick-Marking Bank: Cover this bank with your hand or a sheet of paper to test yourself against Page 2, or use for rapid peer marking.		Answers 1–50	
1. Surgeons moved from 'antiseptic' surgery (killing germs already in the wound) to 'aseptic' surgery (preventing germs from entering the operating theatre in the first place).	✓[X]	2. The steam autoclave (invented in 1881) used to sterilise surgical instruments.	✓[X]
3. 1895	✓[X]	4. The radiation doses were 1,500 times stronger than today, causing severe burns and hair loss.	✓[X]
5. The glass tubes were incredibly fragile, the machines were too heavy to move easily, and taking a single image could take up to 90 minutes.	✓[X]	6. He discovered three blood groups (A, B, and O), meaning doctors could finally match donor and patient blood types to prevent fatal rejection.	✓[X]
7. Blood coagulated (clotted) as soon as it left the body, meaning it could not be stored in a bank. Transfusions had to be done with the donor and patient connected directly by a tube.	✓[X]	8. The flat agricultural land had its drainage systems destroyed by constant artillery, turning the battlefield into a horrific, waterlogged, muddy swamp.	✓[X]
9. A piece of land that juts out into enemy territory, making it highly dangerous because it is vulnerable to attack from three sides.	✓[X]	10. The Battle of the Somme	✓[X]
11. An extensive underground hospital built into the chalk tunnel network underneath the city, sheltering the wounded from artillery fire.	✓[X]	12. The first ever blood bank, using stored blood to treat soldiers suffering from severe shock.	✓[X]
13. The soldiers' wounds were already deeply infected with anaerobic bacteria from the heavily manured, muddy soil the moment they were injured in the trenches.	✓[X]	14. The deep mud, craters, and destroyed roads made motorized transport impossible near the front, forcing stretcher-bearers to manually carry the wounded for miles.	✓[X]
15. They were completely unprepared for the unique environmental illnesses, the massive casualty numbers, and the horrific shrapnel wounds caused by high-explosive artillery.	✓[X]	16. A static, defensive conflict where millions of men lived for months in unhygienic dugouts, exposed to weather, rats, and human waste while trying to wear down the enemy.	✓[X]
17. The deep mud was so thick that wagons had to be pulled by six horses instead of two, and the severe shaking often worsened the patients' injuries.	✓[X]	18. A complex network of zigzag dugouts consisting of a front line, support trench, reserve trench, and communication trenches linking them.	✓[X]
19. 1907	✓[X]	20. High-tech medicine (aseptic surgery, x-rays, blood typing) had emerged, but key limitations in mobility and blood storage meant it was not yet adapted for a muddy, static battlefield.	✓[X]
21. Standing for days in cold, waterlogged mud without changing socks.	✓[X]	22. The feet would swell, blister, turn gangrenous, and often require surgical amputation to save the soldier's life.	✓[X]
23. By ordering soldiers to change their socks twice a day and systematically rub their feet with whale oil.	✓[X]	24. A flu-like illness characterized by severe joint pain, shivering, and high fevers.	✓[X]
25. The millions of body lice living in the seams of the soldiers' unwashed uniforms.	✓[X]	26. They set up massive delousing stations and bathhouses to disinfect clothing with hot steam.	✓[X]
27. A vulnerable 'bulge' in the Allied line surrounded by the enemy on three sides, allowing the Germans to fire down from higher ground.	✓[X]	28. They held onto the salient to stop the Germans advancing to the sea, successfully retaining control of the vital English Channel ports.	✓[X]
29. Chlorine gas	✓[X]	30. Offensive mining to tunnel under and blow up the German positions from below.	✓[X]
31. To relieve extreme military pressure on the French army fighting at Verdun.	✓[X]	32. The British suffered nearly 60,000 casualties in a single day, completely overwhelming the medical services.	✓[X]
33. They dug a 2.5-mile network of tunnels sheltering 25,000 troops, which included a fully functioning underground hospital.	✓[X]	34. It was where troops would retreat to if the frontline came under heavy attack or was overrun.	✓[X]
35. To prevent enemy fire or the blast wave from an exploding artillery shell from travelling straight down the line.	✓[X]	36. It made it incredibly difficult and physically exhausting for stretcher-bearers to safely manoeuvre seriously wounded men around the tight corners while under fire.	✓[X]
37. The heavy motor ambulances frequently got permanently stuck in the deep, waterlogged mud.	✓[X]	38. The delay caused men to die from blood loss or shock, and allowed deep infections like gas gangrene to take hold before they could reach a Casualty Clearing Station.	✓[X]
39. A stomach infection causing severe diarrhea and dehydration, spread by drinking contaminated water from shell holes.	✓[X]	40. That the relentless psychological trauma, noise, and sheer terror of the artillery war caused severe mental breakdowns, resulting in tiredness, nightmares, and uncontrollable shaking.	✓[X]
41. High-explosive artillery shells and shrapnel.	✓[X]	42. The jagged shrapnel tore through flesh, physically dragging pieces of muddy, bacteria-soaked uniform deep into the wound.	✓[X]
43. It was agricultural farmland that had been heavily fertilized with animal manure, packing the soil with the anaerobic bacteria that caused tetanus and gas gangrene.	✓[X]	44. It was a rapid infection that produced a foul-smelling gas inside dying muscle tissue, turning the flesh black and potentially killing a man within a single day.	✓[X]
45. Antiseptics only cleaned the surface of a wound; they could not reach the lethal bacteria that had been driven deep into the muscle by the force of an explosion.	✓[X]	46. By administering routine anti-tetanus injections to wounded soldiers from late 1914 onwards.	✓[X]
47. Soldiers were fighting in trenches; their bodies were protected by earth, but their heads were exposed while wearing only soft cloth caps.	✓[X]	48. The steel Brodie helmet	✓[X]
49. Chlorine Gas	✓[X]	50. It destroyed the respiratory system, causing victims to slowly suffocate as their lungs filled with fluid.	✓[X]

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Official Mark Scheme & Knowledge Vault Answers · Answers 51 to 100 (Self & Peer Marking Bank)

MARK SCHEME 2

🔦 Quick-Marking Bank: Use for rapid recall checking against Vault Part 2 (Page 3). Log total correct recall items on Page 1.		Answers 51–100	
51. They urinated on cotton pads or handkerchiefs and pressed them to their faces to chemically neutralize the chlorine.	[✓][X]	52. Phosgene Gas	[✓][X]
53. It was completely odourless and worked slowly, causing severe internal and external blisters that could burn skin straight through uniforms.	[✓][X]	54. The rapid development and distribution of highly effective gas masks successfully protected the vast majority of troops.	[✓][X]
55. Trench Fever	[✓][X]	56. By ordering soldiers to change their socks twice a day and systematically rub their feet with whale oil.	[✓][X]
57. A poorly understood psychological trauma (PTSD) caused by the constant bombardment of war, resulting in nightmares, loss of speech, and uncontrollable shaking.	[✓][X]	58. Craiglockhart Hospital	[✓][X]
59. NYD.N (Not Yet Diagnosed, Nervous)	[✓][X]	60. Because the harsh conditions caused so many massive casualties from illnesses (like trench foot and fever) that they had to actively prevent disease just to keep the army fit enough to fight.	[✓][X]
61. To provide a highly organised, multi-stage system of medical triage to rescue, transport, and treat massive numbers of casualties without overwhelming frontline units.	[✓][X]	62. Teams of four to six stretcher-bearers, who carried the casualties by hand through the mud.	[✓][X]
63. Within 200m of the frontline, often in a communication trench or dugout.	[✓][X]	64. They could only provide immediate first aid (like bandages and tourniquets) to get the walking wounded back to the fight or prepare serious cases for transport; they could not perform surgery.	[✓][X]
65. Units of the Field Ambulance, consisting of medical officers, orderlies, and eventually nurses.	[✓][X]	66. The Times newspaper launched a public appeal in 1914, raising money to purchase 512 new motor ambulances.	[✓][X]
67. Because heavy motor vehicles frequently got bogged down and entirely stuck in the deep, waterlogged mud of the battlefield.	[✓][X]	68. On specially designed ambulance trains and canal barges.	[✓][X]
69. 7 to 12 miles back from the frontline, safely out of artillery range but positioned near railway lines for rapid transport.	[✓][X]	70. Triage, where patients were sorted into three groups: the walking wounded, those needing hospital treatment, and those with no chance of recovery.	[✓][X]
71. It took over the role of performing critical, life-saving surgery (like amputations) from the Base Hospitals to stop the infection before transport.	[✓][X]	72. Near the French and Belgian coastal ports, ready to ship patients back to 'Blighty' (Britain).	[✓][X]
73. The longer-term recovery of patients and the development of specialist wards for complex issues like head wounds or gas poisoning.	[✓][X]	74. From 9,000 men to 113,000 men.	[✓][X]
75. Queen Alexandra's Imperial Military Nursing Service, a corps of highly trained professional military nurses.	[✓][X]	76. First Aid Nursing Yeomanry.	[✓][X]
77. They drove the motor ambulances, navigating terrible roads and deep mud under active shellfire to transport the wounded.	[✓][X]	78. They transported supplies, set up cinemas, and operated mobile bath units that could bathe up to 40 men an hour.	[✓][X]
79. It was a massive, fully functioning underground hospital built into existing chalk tunnels beneath the town of Arras.	[✓][X]	80. Because it functioned as a fully equipped facility incredibly close to the frontline (with 700 stretcher beds, electricity, and water), while remaining completely safe from enemy artillery.	[✓][X]
81. Because the chaotic, muddy casualty stations were impossible to keep perfectly sterile, and traditional carbolic acid did not kill gas gangrene bacteria effectively.	[✓][X]	82. Aggressively and rapidly cutting away all dead, damaged, and infected tissue from around a wound to physically remove the gangrene before stitching it up.	[✓][X]
83. A system where tubes pumped a chemical sterilized salt solution directly into deep wounds to continually wash out and kill the infection-causing bacteria.	[✓][X]	84. The solution only stayed fresh for six hours and had to be constantly made as it was needed.	[✓][X]
85. Immediate amputation of the infected limb (resulting in over 240,000 lost limbs by 1918).	[✓][X]	86. 80%	[✓][X]
87. Because traditional splints did not keep the leg straight, meaning the jagged broken bones ground together, severing major arteries and causing fatal internal bleeding and shock.	[✓][X]	88. Hugh Owen Thomas	[✓][X]
89. It rigidly pulled the leg lengthways (traction), completely stopping the broken bones from grinding on each other and preventing fatal blood loss and shock during transport.	[✓][X]	90. X-Rays	[✓][X]
91. Six mobile X-ray units (vans) were developed to operate in the British sector, allowing them to be driven directly to Casualty Clearing Stations.	[✓][X]	92. The delicate glass tubes overheated very quickly, meaning the machines could only be used for an hour before needing to cool down.	[✓][X]
93. Transfusions had to be done 'arm-to-arm', directly transferring blood from a live donor to the patient before it clotted.	[✓][X]	94. Sodium Citrate	[✓][X]
95. They added citrate glucose to the blood, which allowed it to be safely stored for up to four weeks.	[✓][X]	96. Oswald Hope Robertson	[✓][X]
97. He stored the blood in glass bottles packed with ice and sawdust.	[✓][X]	98. Harold Gillies	[✓][X]
99. Because traditional peacetime surgical methods completely failed to treat the complex, infected shrapnel wounds, forcing doctors to innovate radical new techniques out of sheer desperation.	[✓][X]	100. They could not detect pieces of dirty clothing that had been driven into wounds by shrapnel, which often caused lethal infections.	[✓][X]

SECTION A: THE HISTORIC ENVIRONMENT

Round 1: The Stepped Ladder · Question 1 (Features) & Question 2(b) (Follow-Up Investigation)

ROUND 1: STEMS 1 & 2B

Question 1 (a): Feature Question (2 Marks)

F-D Formula · ~3 Mins

Describe one feature of the work of Casualty Clearing Stations (CCS) on the Western Front.

 **Level 1 Launchpad:** Starter: "One feature of Casualty Clearing Stations was that..."
Facts to Include: Located 7–12 miles back near railways; triage (walking wounded, urgent surgery, dying); performed critical amputations and chest surgeries before base evacuation.

One feature of Casualty Clearing Stations was that...

Specifically, (add one precise historical detail or statistic)...

Question 1 (b): Feature Question (2 Marks)

F-D Formula · ~3 Mins

Describe one feature of the use of the Thomas splint on the Western Front.

 **Level 1 Launchpad:** Starter: "One feature of the Thomas splint was that..."
Facts to Include: Introduced in 1915 by Robert Jones; rigid frame pulled leg straight to stop bone ends grinding; dramatically dropped compound fracture mortality from 80% to 20%.

One feature of the Thomas splint was that...

Specifically, (add one precise historical detail or statistic)...

Question 2 (b): Follow-Up Investigation Grid (4 Marks)

4-Step Enquiry · ~6 Mins

Study Source B (Edith Smith letter: "The pipes freeze entirely, meaning we must strictly ration the water provided to the wounded."). How could you follow up Source B to find out more about the severe conditions faced by medical staff? Complete the table below.

Detail in Source B that I would follow up:	"The pipes freeze entirely, meaning we must strictly ration the water provided to the wounded." What specific impact did water shortages have on surgical sterilization and infection rates in winter 1917?
Question I would ask:	Base Hospital Daily Medical Superintendent Logbooks and Sanitary Inspection Reports (1917–18).
What type of source I would use:	It would provide official daily records of sanitation breakdowns and hospital infection rates during freezing months.
How this source would help me answer my question:	

Question 2 (a): Source Utility (8 Marks)

C-O-P Formula · ~12 Mins

Study Sources A and B. How useful are Sources A and B for an historian studying the immense challenges of treating casualties on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context.

Source A (Chaplain Arthur Davies, Somme CCS, July 1916):

"The casualties have started to arrive... The volume of patients is overwhelming; every bed is occupied, and men are fortunate just to find a spot on the bare ground. Many will pass away before surgery is possible. We have admitted 1,500 casualties, and the stream has not stopped."

Source B (Edith Smith, VAD Nurse, Base Hospital, Dec 1917):

"Our morning duties begin at 3.30 am! The freezing temperatures are unbearable. Icicles hang thick over ward windows. Even kettles, rubber hot water bottles, and sponges have frozen solid. The pipes freeze entirely, meaning we must strictly ration water provided to the wounded."

PROVENANCE CLUES (Author, Audience, Motive): Who wrote each source (eyewitness chaplain at CCS vs frontline VAD nurse at base hospital)? What was their motive (private diary recording sheer volume vs private letter home detailing physical conditions)? How does their perspective make the sources useful despite subjective emotional focus?

PARAGRAPH 1: UTILITY OF SOURCE A (CONTENT + PROVENANCE + CONTEXT)

Source A is useful for investigating casualty challenges because the content reveals...

From my own knowledge, this accurately reflects the Somme offensive because...

Furthermore, the provenance makes it useful because as a private chaplain's diary...

PARAGRAPH 2: UTILITY OF SOURCE B & COMPARATIVE JUDGEMENT (CONTENT + PROVENANCE + CONTEXT)

Source B is also useful because it highlights the severe environmental difficulties...

Specifically, my own knowledge confirms that Base Hospitals in winter 1917...

The provenance enhances its utility because as a private letter from a VAD nurse...

Overall, both sources are mutually useful because together they demonstrate...

Level 3 (Grade 9) Utility Glue: Weigh up typicality vs limitation — note that Source A shows peak crisis conditions on 1 July 1916 (57,000 casualties in one day), while Source B reflects seasonal winter extremes at the coast.

SECTION A: THE HISTORIC ENVIRONMENT

Questions 1(a) & 1(b): Features (2 + 2 = 4 Marks)

Independent Toolkit

Fact Vault (AO1)

- **Mobile X-rays:** 6 mobile units in British sector; located at CCS; couldn't detect clothing fragments; tubes overheated.
- **Trench Fever:** Caused by body lice; flu-like pain in legs; affected ~15% of men; led to delousing stations and bathhouses.

⚠ **Trap:** Do not confuse Trench Fever (lice) with Trench Foot (waterlogged mud/cold).

1(a) Describe one feature of mobile X-ray units near the frontline:

One feature of mobile X-ray units was...

1(b) Describe one feature of trench fever:

One feature of trench fever was...

Question 2(b): Follow-Up Practice (4 Marks)

Historical Records Directory

Study this extract from an RAMC officer at Ypres: "The chlorine gas casualties stumbled in clutching their throats. We washed their blinded eyes with bicarbonate of soda solutions, but many died from asphyxiation."

Detail in Source to follow up:	"We washed their blinded eyes with bicarbonate of soda solutions..."
Question I would ask:	How effective were early chemical antidotes compared to later respirator gas masks? RAMC Field Ambulance Casualty Admission Books & War Office Gas Defence Bulletins
What type of source I would use:	(1915). It would record official medical survival rates of gas victims before and after the issue of
How this source would help me:	Small Box Respirators.

Question 2(a): Source Utility Independent Execution (8 Marks)

Wartime Trauma & Gas Warfare

How useful are Source C (RAMC Logbook, 1915) and Source D (Official Training Manual on Trench Foot, 1916) for an enquiry into the prevention of trench illness on the Western Front?

🕒 Provenance Evaluation Matrix

- 📄 Official Reports: High factual accuracy regarding procedures; may minimize failure to maintain morale.
- 👨👩 Doctor/Nurse Accounts: Direct clinical insight into actual suffering; localized to one specific sector.

🇬🇧 Causal & Utility Connectives

- "The content is highly useful because it reveals..."
- "The provenance strengthens its value because..."
- "However, the source is limited in scope because..."

🎯 Level 3 Standard: Explicitly evaluate how the author's professional role and purpose shape what was included or omitted.

PARAGRAPH 1: UTILITY OF SOURCE C

Source C is useful for an enquiry into prevention because...

Its provenance as an RAMC logbook means...

PARAGRAPH 2: UTILITY OF SOURCE D & SYNTHESIS

Source D is useful in a different way because it shows official policy on...

However, an official manual may not reflect actual frontline reality because...

Comparing both sources, an historian learns that...

SECTION A: THE HISTORIC ENVIRONMENT

1. Question 1 Features Selector:

Exam Prompt	Feature 1 (Core Concept)	Supporting Specific Fact / Detail (AO1)
Q1(a) Underground Hospital at Arras	Built into chalk quarries; fully equipped	700 beds, running water, electricity, operating theatre, mortuary.
Q1(b) Blood Transfusion at Cambrai	First large-scale indirect blood bank (1917)	Oswald Robertson used sodium citrate/dextrose; treated 20 severely wounded men.

2. Question 2(a) Source Utility Planning Grid:

Source	Content Knowledge (What it reveals)	Provenance Analysis (NOP & Evaluation)
Source A (Medical)	Specific wound types (shrapnel, compound fracture, gangrene)	Doctor/nurse eyewitness account; high clinical validity but localized.
Source B (Soldier)	Stretcher bearer delays in mud, shell craters, shell shock	Infantry diary; emotional impact; subject to wartime censorship.

3. Question 2(b) Follow-Up Pre-Flight Selector:

Detail to Quote	Targeted Question	Authentic Source Type	How it Answers
Exact phrase about delays in stretcher evacuation	What was the average time taken from RAP to CCS in the Ypres salient?	RAMC Field Ambulance Section War Diaries (WO 95)	Provides recorded times of stretcher bearer transfers across muddy terrain.

 **PRE-FLIGHT AUDIT CHECKLIST:** ☐ Did I state TWO separate features for Q1? ☐ Did I analyze BOTH content and provenance for Sources A & B? ☐ Is my Q2(b) source a specific historical record (not 'the internet')?

SECTION A: THE HISTORIC ENVIRONMENT

Round 3: The Exam Pitch · Authentic Edexcel Section A Simulation (16 Marks · 25 Mins)

TIMED EXAM PITCH

AO1 Knowledge: Precise names, locations (Ypres, Somme, Cambrai), equipment, and dates.

AO3 Sources: Balanced analysis of utility using content, contextual knowledge, and provenance.

Question 1(a) (2 Marks):Describe one feature of the underground hospital at Arras: _____

Question 1(b) (2 Marks):Describe one feature of blood transfusion techniques on the Western Front: _____

Question 2(a) (8 Marks):Source A is useful because _____

Source B is also useful because _____

Question 2(b) (4 Marks): Complete the follow-up investigation table below:

Detail in Source B to follow up:	_____
Question I would ask:	_____
What type of source I would use:	_____
How this source would help me:	_____

TOP 3 FATAL EXAMINER TRAPS TO AVOID FOR SECTION A

• The "Generic Source" Trap

Never write "I would check a diary" or "the internet" in Q2(b). You must specify an authentic record type, like RAMC War Diaries or Hospital Admission Registers.

• The "Treachery of Bias" Trap

Never reject a source simply because "it is biased so it's useless". Sources written with strong emotion or wartime censorship are highly useful for showing attitudes!

• The Single Feature Trap

Remember Q1 is split into two questions: 1(a) and 1(b). Do not combine them into one paragraph. Each requires one identified feature + one specific factual detail.

SPECIFICATION PRACTICE BANK: 100% CURRICULUM COVERAGE

Every remaining Western Front specification bullet point tested below

- 1. Features (2m):** Describe one feature of the role of the FANY (First Aid Nursing Yeomanry) in ambulance transport.
- 2. Features (2m):** Describe one feature of the effects of mustard gas on soldiers.
- 3. Source Utility (8m):** How useful are official trench casualty statistics compared to private letters for studying the effectiveness of Brodie helmets?
- 4. Features (2m):** Describe one feature of the Carrel-Dakin method for treating gas gangrene.
- 5. Follow-Up (4m):** Detail: "*Ambulance barges along the Somme-Yser canal were fitted with spring beds.*" Formulate a question and name a primary source.

⚡ **QUICK-FIRE FOLLOW-UP SOURCE DRILL:** When asked to research: (1) *Wound infections* → RAMC Medical Officer Daily Logs; (2) *Frontline delousing routines* → Battalion War Diaries; (3) *Blood transfusions* → CCS Surgical Register.